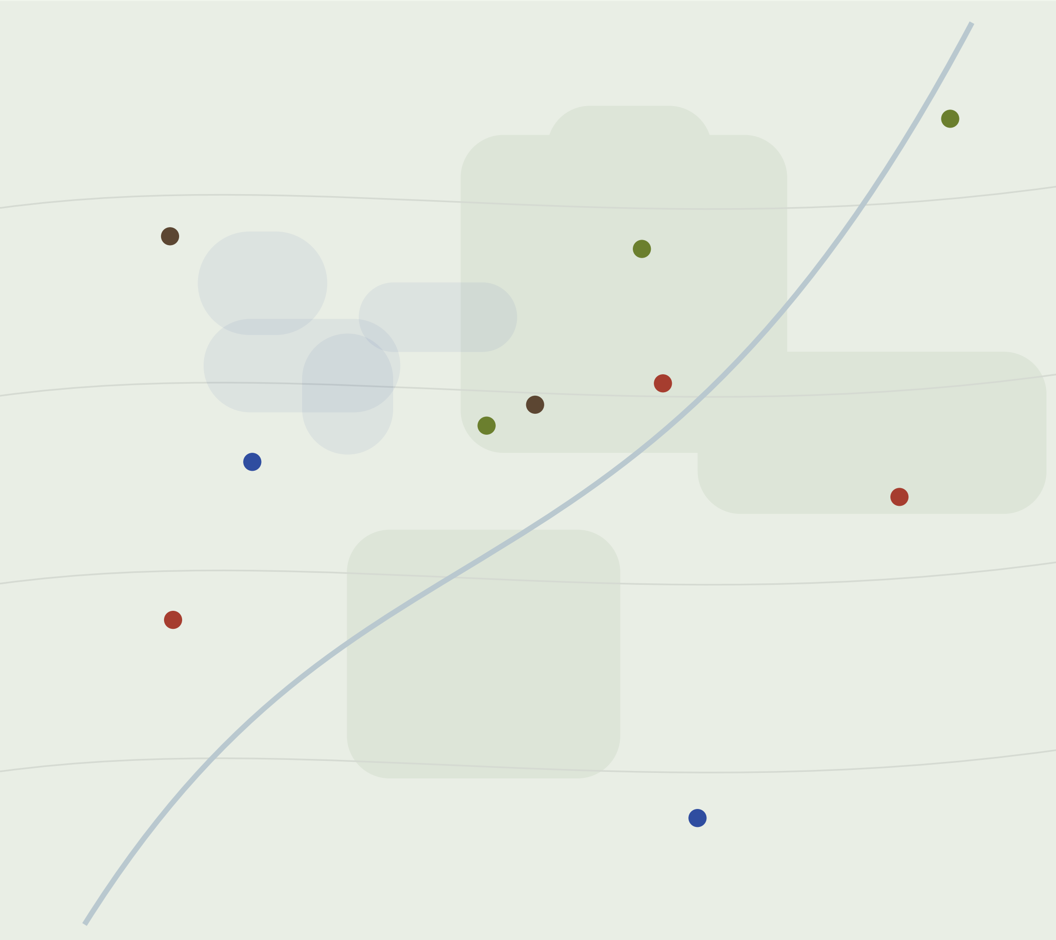
The Living Map -- Round 4

Field Desk bones . Pigment Index skin . Night Survey after dark . the almanac's conditions, live



SAME PLACE, 2 PM

SAME PLACE, 11 PM

ROUND 4 / WHAT YOU CHOSE, RESOLVED

One site from three rounds of parts

UX SKELETON	**Field Desk (R3-A)**	map dominant; search, cards, recipes float above it
LIGHT-MODE SKIN	**Pigment Index (R2-III)**	material-derived color, mono labels, one fixed display face -- no seasonal type changes
DARK REGISTER	**Night Survey (R2-I)**	not a permanent mode: the night state of a time-of-day system
BOLDNESS DIAL	**Overprint (R2-II)**	its poster energy offered at three volumes for the floating cards
NEW LAYER	**Living conditions**	weather, sun and moon, tide -- the almanac's own subjects, drawn on the map

The argument of this round: an almanac is a book of conditions -- weather, daylight, tides, the moon -- consulted because it knows what today is. A map that shows where things grow but not what the sky is doing is only half an almanac. The conditions layer closes that gap, gives the map its life, and gives night mode a reason to exist beyond taste.

Constants, as always: occurrence is not permission, in every popup. Safety colors fixed and CVD-safe in every register. Conditions inform; they never promise. Any new data source lands in ATTRIBUTION.md with license notes before it ships.

RESEARCH / THE DATA IS FREE

Every condition has a public, key-less source

CONDITION	SOURCE	TERMS	STRATEGY
Weather now + forecast	NWS api.weather.gov	free, no key, no account; alerts, observations, forecasts	`fetch per viewport center, cache 30 min`
Radar-style precipitation	Open-Meteo	open-source, free non-commercial, no key; 15+ national models	`hourly precip grid; cache 30 min`
Recent rain (fungi logic)	Open-Meteo historical	past-72h precipitation by point	`drives 'flush likely' pulses; cache 6 h`
Tides + water level	NOAA CO-OPS API	public, free, no enforced limits; predictions + live levels	`nearest station per viewport; cache 1 h`
Sun + moon	computed client-side	solar position, twilight times, moon phase -- SunCalc-style math, ~2 KB	`zero network, exact for user's location`
Map light itself	Mapbox Standard lightPreset	dawn / day / dusk / night presets, one setConfigProperty call	`or two custom styles if we stay on classic`
Shellfish biotoxin closures	state health depts	hand-encoded like NPS compendiums; permission-required labeling	`updated by the 4am research loop`

Graceful degradation is the contract: if every API is down, the site is exactly today's site -- conditions are an enhancement layer, never a dependency. All sources enter ATTRIBUTION.md with license notes before launch.

WHY CONDITIONS BELONG

Conditions gate the gathering -- they are not decoration

CONDITION	WHAT IT MEANS FOR CRAFT + FORAGING	WHAT IT MEANS FOR SAFETY
Rain, last 72 h	fungi flush 2-3 days after; inky caps overnight; lichens rehydrate	slippery banks; swollen creeks
First frost	walnut hulls drop; persimmons sweeten; sap lines wake in late winter	ice underfoot; season's hard stop
Wind	seed and fiber gathering windows; drying days for harvested material	widow-makers in the canopy -- stay out of old stands
Sun + twilight	golden hour for color-true berry picking; UV and heat windows	legal gathering hours in many parks end at dusk
Tide	intertidal seaweed and shellfish access opens twice a day	biotoxin closures override everything; turning tides strand
Moon	night walks for evening primrose, moths, low-tide nights	headlamp ethics; private-land lines invisible after dark

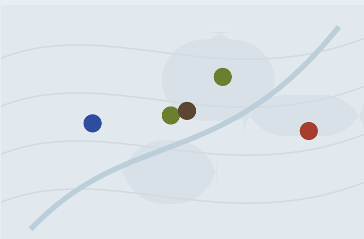
Design rule that follows: every condition drawn on the map must answer 'so what should I do?' -- a rain wash without the fungi pulse is weather TV; with it, it is an almanac. And conditions never promise: 'flush likely' is a likelihood, said so.

THE FOUR LIGHT REGISTERS

The site keeps the sun's hours

DAWN

CIVIL DAWN TO SUNRISE +1H



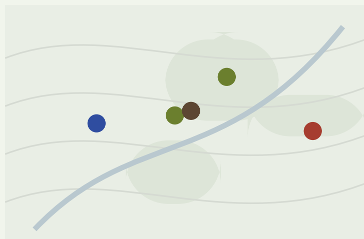
Elderberry

RIPE WK 24-28

ALLOWED HERE

DAY

THE WORKING REGISTER



Elderberry

RIPE WK 24-28

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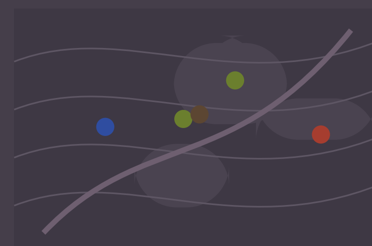
DUSK

GOLDEN HOUR TO CIVIL DUSK



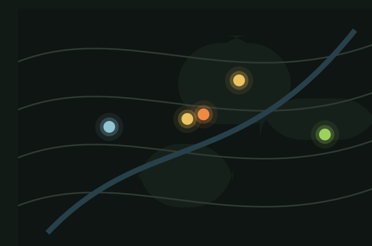
Elderberry

RIPE WK 24-28

ALLOWED HERE

NIGHT

DUSK TO DAWN



Elderberry

RIPE WK 24-28

ALLOWED HERE

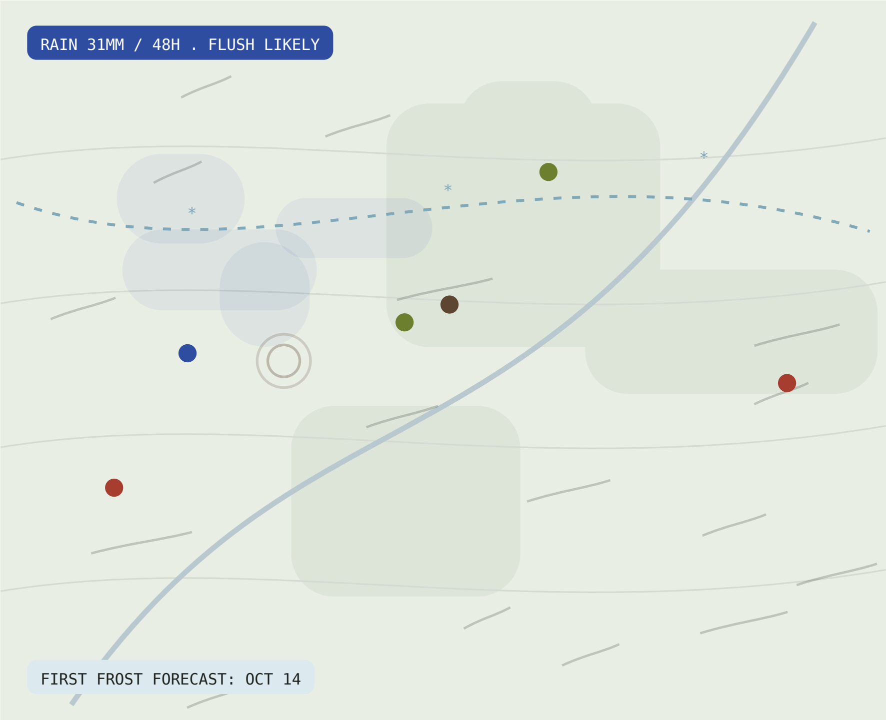
Mechanics: solar position is computed client-side, so the register follows the user's actual sky -- sunset in Maine is not sunset in Texas. Each register is one CSS variable set plus one Mapbox lightPreset call; transitions crossfade over a minute, never mid-task. A manual override (sun/moon toggle) always wins and is remembered.

Safety palettes are tuned per register (luminance-shifted, hue-stable) and checked for CVD in all four. Dusk and dawn are brief, atmospheric, and optional -- classrooms can pin DAY; the projector never surprises anyone.

CONDITION LAYER

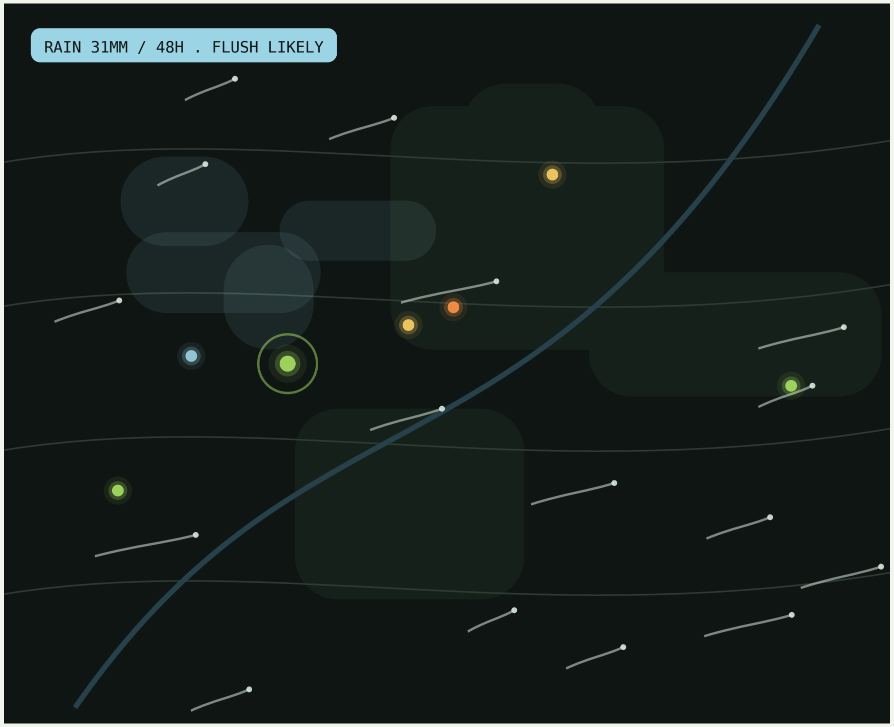
Weather: rain remembered, wind drawn, frost foretold

Open-Meteo + NWS; the map recalls the last three days because mushrooms do.



DAY REGISTER

Day: precipitation as quiet pigment-indigo washes; wind as faint drifting strokes; the first-frost forecast is a dotted isoline with a date. Rain memory rings pulse around fungal records when the past-72h total crosses a species' flush threshold.



NIGHT REGISTER

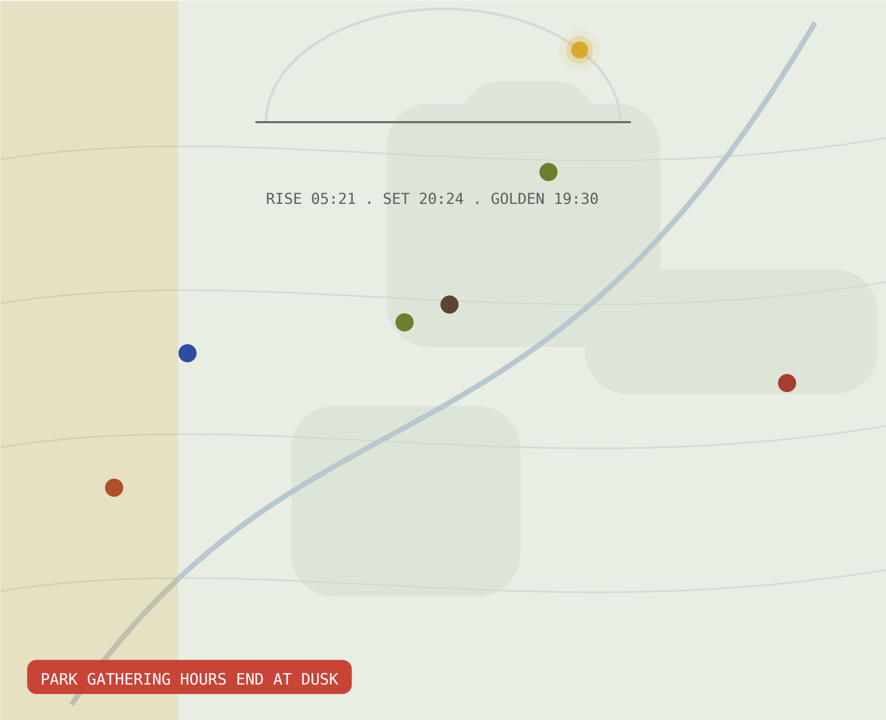
Night: the same data becomes light -- washes luminesce faintly, wind streaks carry glowing heads (the firefly grammar), pulses brighten. Nothing new is added; the register only re-voices it.

The almanac move: weather is not a forecast widget in a corner -- it is drawn where it falls, and it answers 'so what': flush pulses, frost dates, drying days.

CONDITION LAYER

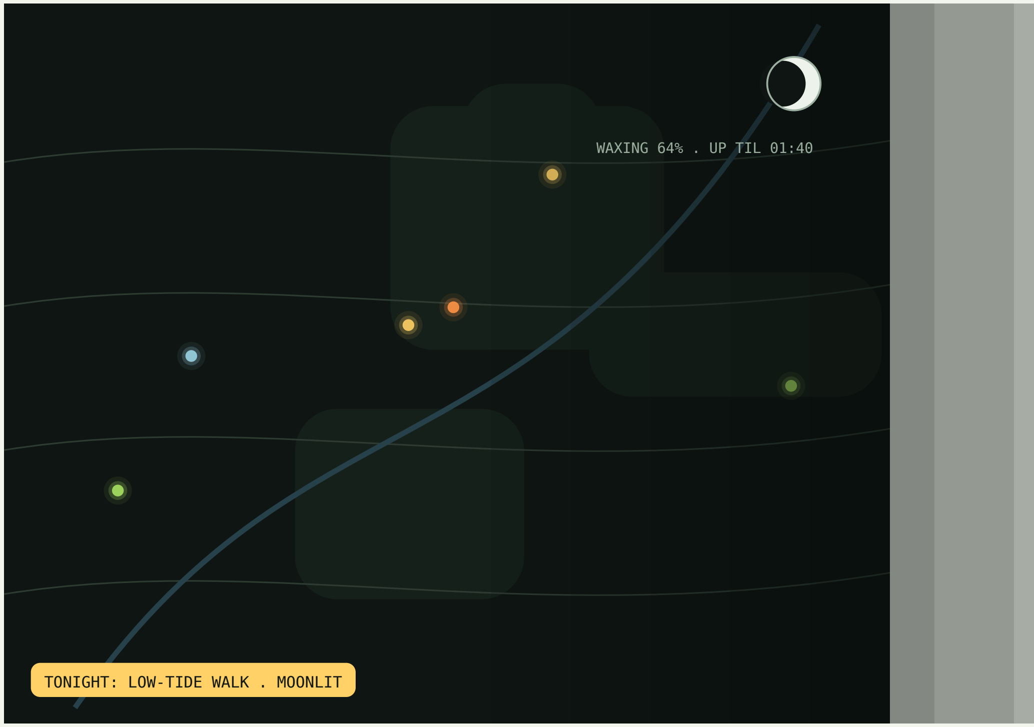
Sun + moon: the day's edges, kept by the site

Computed client-side -- zero network. The scrubber gains the sun's arc.



DAY REGISTER

Day: a sun-path arc rides the season scrubber with rise/set/golden-hour ticks; the golden band tints the map's west edge as it approaches. Where park rules end at dusk, the countdown chip says so plainly -- a rule, not a mood.



NIGHT REGISTER

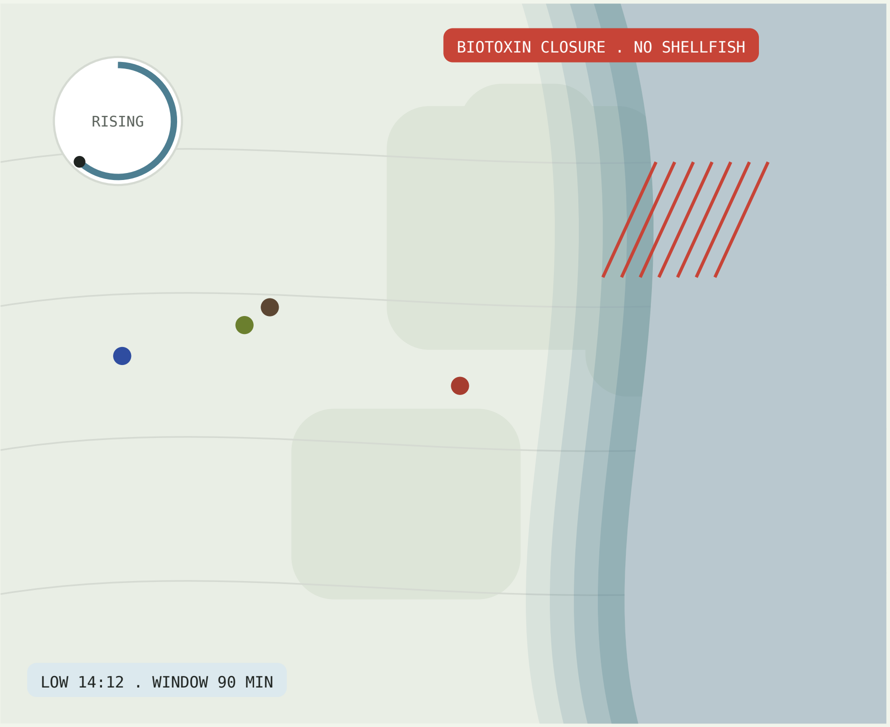
Night: the terminator becomes visible geography -- the dark drawn as deepening ink across the map. The moon chip carries phase, illumination, and set time: the almanac's oldest data, doing modern wayfinding.

Time-of-day registers and this layer are one system: the map's light preset follows the same solar math the widgets display. The site does not have a dark mode; it has a sky.

CONDITION LAYER

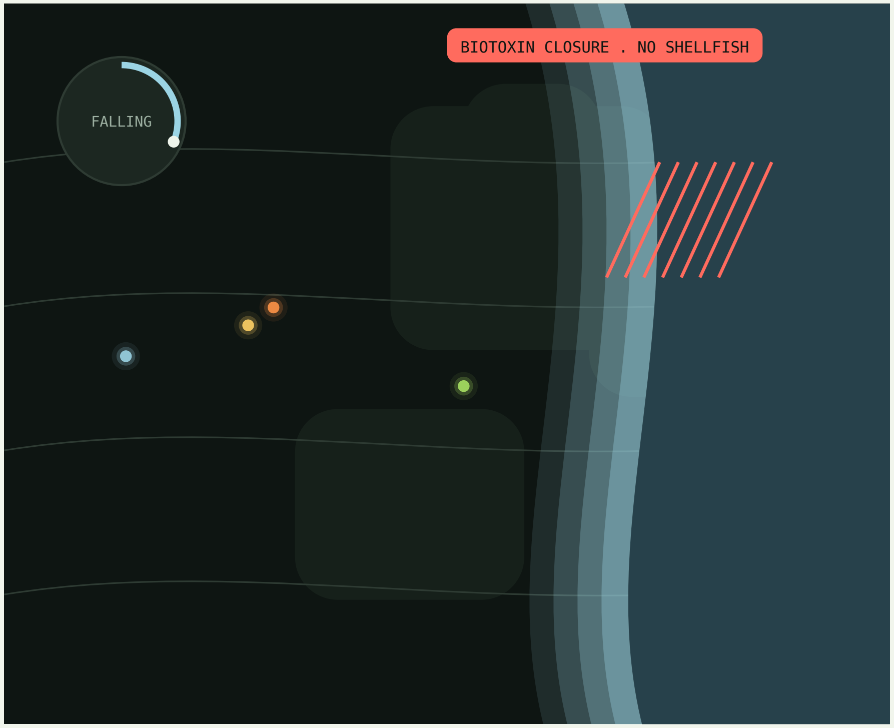
Tide: the intertidal breathes twice a day

NOAA CO-OPS predictions; the shore is the only map edge that moves.



DAY REGISTER

Day: stepped bands show the zone the falling tide is opening; the tide clock is a 12.4-hour dial, rising or falling. The low-tide window -- the almanac's gift to seaweed and clam gatherers -- is a chip with minutes, not poetry.



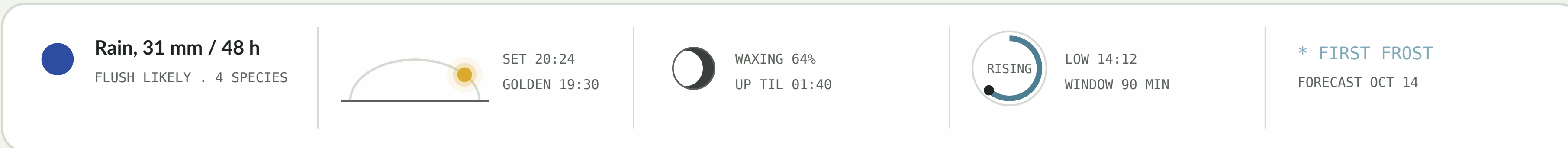
NIGHT REGISTER

Night: bands carry moonlit luminance (low-tide nights are real gathering events); the dial inverts. Biotoxin closures are the one mark that never changes register logic: maximum-contrast hatch + chip, day or night.

Closures override everything, everywhere, always -- the layer exists as much to say 'not this week' as 'now'. Inland users simply never see this layer; it mounts only when the viewport touches a coast.

THE CONDITIONS RAIL

One floating strip holds the sky



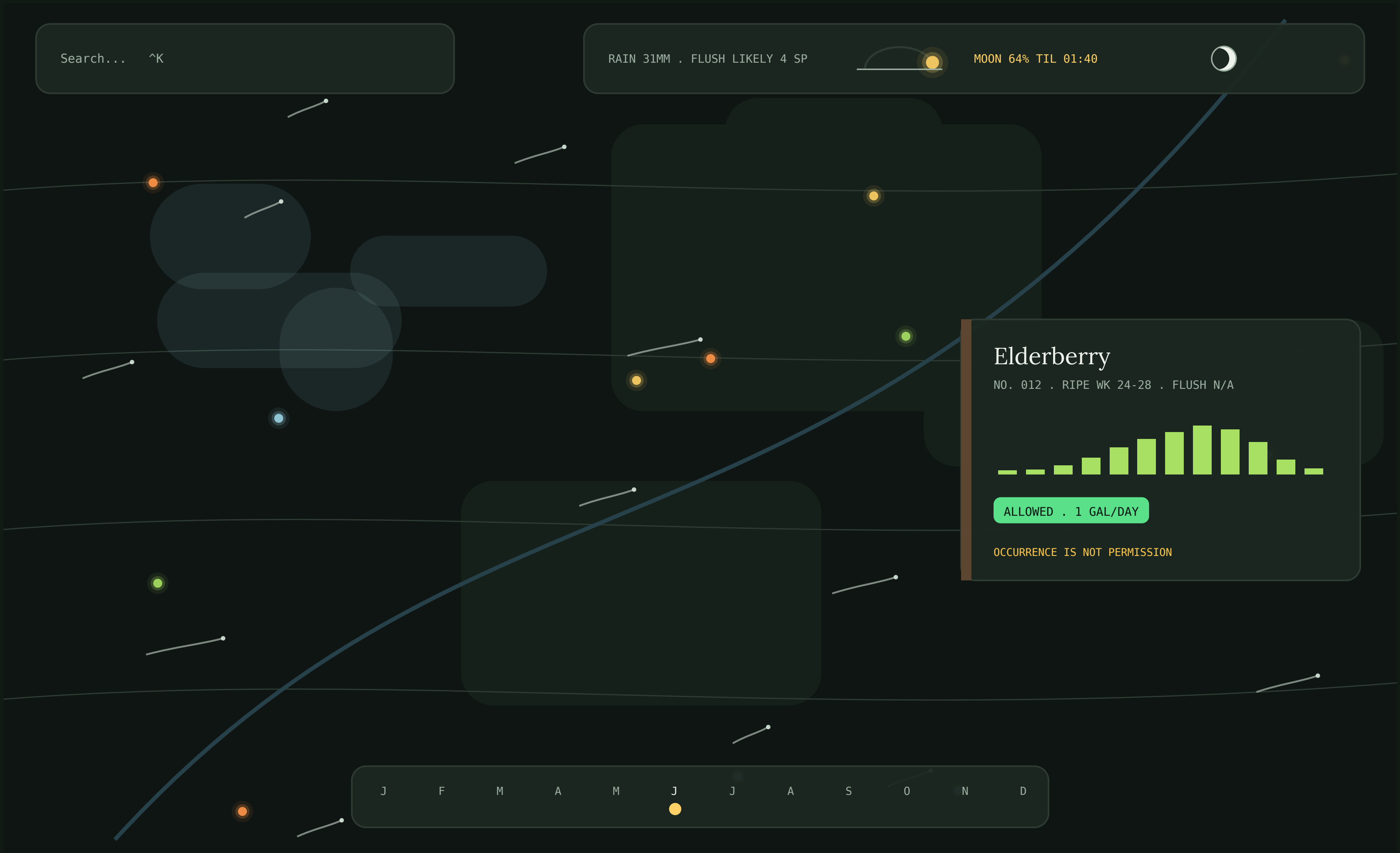
DOCKS TO THE MAP'S TOP EDGE . EACH SEGMENT EXPANDS TO ITS LAYER . COASTAL SEGMENT MOUNTS ONLY NEAR SHORE

Tap a segment and its layer wakes on the map (the others rest -- one condition at a time keeps the map readable). The rail is also where conditions talk to data: 'flush likely . 4 species' is a filter -- tap it and the map shows exactly those four, pulsing where the rain fell.

On phones the rail collapses to a single 'sky chip' (icon + the one most decision-relevant fact); it expands to the full strip on tap. In class, the rail is a lesson plan: today's entry, generated from the same numbers.

Restraint contract: at most one ambient layer animating by default (wind OR rain wash), pulses decay after a minute of stillness, and prefers-reduced-motion stills everything. The map must remain a tool first, weather a voice second.

Everything chosen, together, at 2 pm: Pigment Index ground and cards, walnut spine-tabs, the conditions rail docked above, rain washes where it rained, the scrubber keeping the season. The map dominates; the UI floats; nothing scrolls.

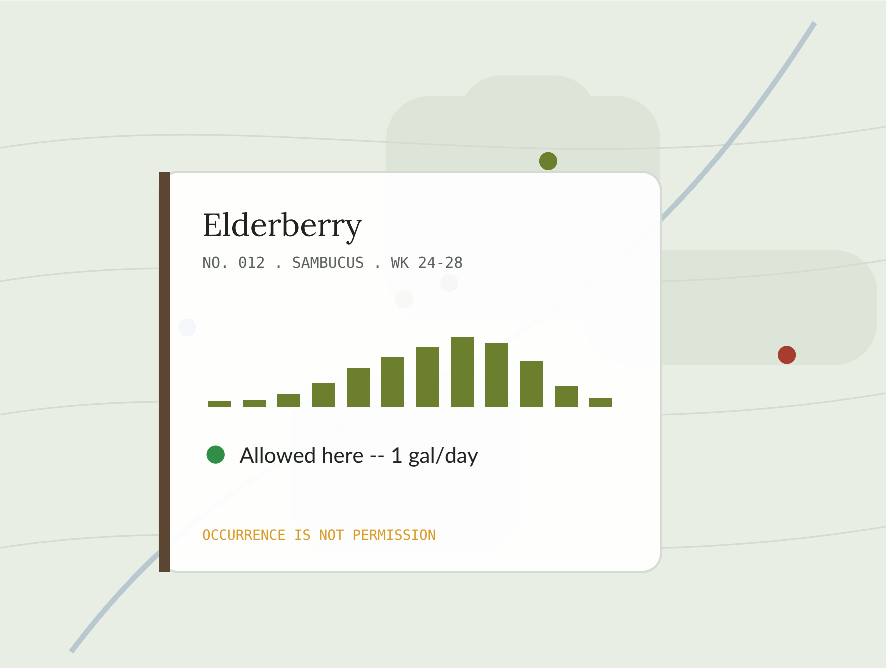


The same view at 11 pm, untouched by hand: the register followed the sun. Data becomes light (Night Survey's grammar), wind carries glowing heads, the moon takes the rail. Safety chips hold their meaning at night luminance.

CARD BOLDNESS / OPTION 1

Quiet Pigment

Round 2-III untouched: hairline shell, pigment spine-tab, serif name, mono data. The most archival voice; identity carried by material color alone.



ON THE DAY MAP

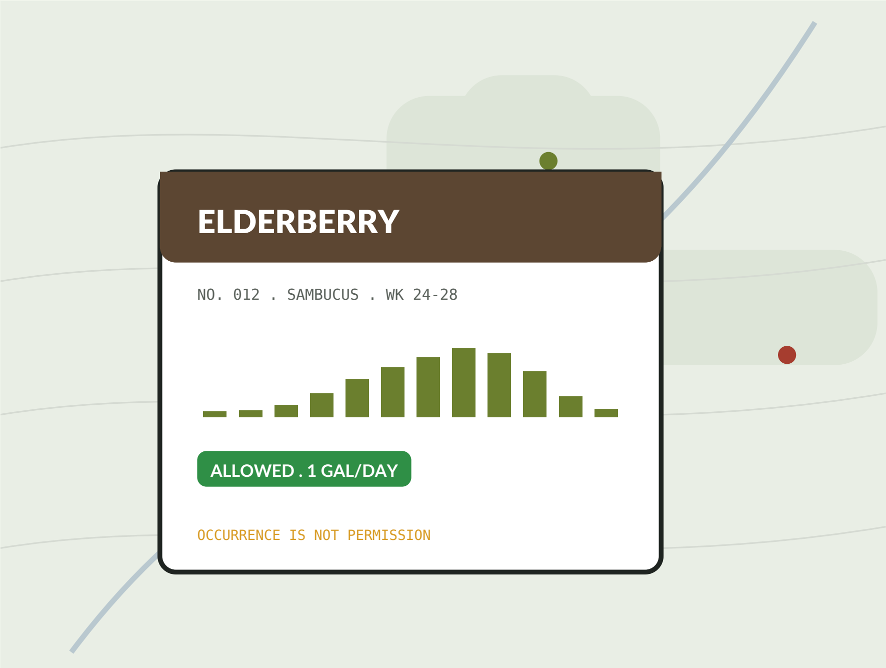
ON THE NIGHT MAP

Strong on reading surfaces; on the living map it can recede -- over weather washes the hairline shell asks the eye to work.

CARD BOLDNESS / OPTION 2

Bold Edition

Pigment Index materials with Overprint's spine: 2 pt ink border, pigment header band with reversed wood-type name, solid chips. No misregistration -- confidence without costume.



ON THE DAY MAP

ON THE NIGHT MAP

The recommended floating-UI voice: holds its edge over rain, wind, and night glow in both registers, and the pigment band still does the teaching.

CARD BOLDNESS / OPTION 3

Poster Overprint

The full print-shop: thick border, offset two-ink shadow, stacked overprint title, halftone corner. Round 2-II's energy, contained to a card.

ON THE DAY MAP

ON THE NIGHT MAP

Joyful and unmistakable -- and a lot, multiplied across a busy map. Earns its keep on recipe broadsides and posters; as the default card it fights the weather for attention.

CARD BOLDNESS / RECOMMENDATION

A dial per surface, not one volume for the site

Floating peek cards + popups	**2 . BOLD EDITION**	the living map is busy; cards need ink-edged authority to float over weather without shouting
Plant pages, rules, citations	**1 . QUIET PIGMENT**	reading surfaces; the spine-tab and pigment fields carry identity, type stays calm
Recipe broadsides + posters	**3 . POSTER OVERPRINT**	the print-shop register earns its volume where instructions want to be pinned to a wall
Safety chips + closures	**FIXED**	identical at every volume, every register -- boldness never applies to meaning
The conditions rail	**2 . BOLD EDITION**	matches the cards it lives among; its data stays mono and quiet

If you want one answer: Bold Edition is the default voice of the floating UI -- Pigment Index's materials with Overprint's spine. Quiet and Poster are its inside voice and outside voice. All three share one geometry, so the dial is a CSS class, not a redesign.

TECH, PERFORMANCE + RISK

Living, within the no-build constraint

REGISTERS	4 CSS variable sets + Mapbox lightPreset; solar-keyed, crossfade 60 s, manual override persists
RAIN WASH	SVG blobs from Open-Meteo grid, CSS opacity; redraw on pan-end only
WIND	<= 120 streak particles, one canvas layer, requestAnimationFrame; killed off-screen and on reduced-motion
PULSES	CSS keyframe rings on existing point layer; decay after 60 s idle
TIDE	two shoreline polygons interpolated by CO-OPS prediction; 1 fetch/hour
CACHING	all condition fetches cached in localStorage with TTL; offline = last known + timestamp
BUDGET	conditions layer <= 60 KB JS total, zero new fonts, zero build steps

RISKS

API drift or outage (degrade to today's site, show data age); motion distracting from the tool (one ambient layer max, idle decay, reduced-motion); battery on phones in the field (particles off on battery-saver); register changing mid-demo in class (pin DAY, remembered per device); condition data implying promises ('likely' language audited the way rules language already is).

ROUND 4 / DECISIONS WE NEED

Five asks

01 **Confirm the spine.**
Field Desk + Pigment Index day + Night Survey night + conditions layer. If yes, this becomes the brief's 'decided foundations' update and Round 5 is a live prototype.

02 **Pick the card volume.**
Quiet / Bold Edition / Poster for the floating UI -- or accept the per-surface dial as recommended.

03 **Scope the first conditions.**
Suggested build order: sun/moon (free, offline, powers registers) -> rain memory + flush pulses -> wind -> tide (coastal viewports only). Trim or reorder freely.

04 **Registers: four or two?**
Dawn and dusk are atmosphere; day and night are function. Shipping two first is simpler; four is more almanac. Your call on ambition.

05 **Name the prototype place.**
A real viewport (your teaching region?) for the prototype so tide/weather demos use true data.

Round 5 proposal: a one-page live prototype on design/relaunch -- real map, real sun math, one condition (rain memory), Bold Edition cards, day/night registers.